

Media Sentiment vs. Economic Reality

A knowledge graph approach to mapping media-driven economic narratives against official indicators across Germany, France, and the Netherlands — enabling proactive risk assessment for European insurance markets.

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Income Protection · Property Business Insurance

Countries: DE · FR · NL · Topic(s): Inflation · Housing · Employment · Period(s): 2021 · 2023

RESEARCH QUESTIONS

VOLUME & SENTIMENT
How does the volume and sentiment of media coverage on inflation, housing, and employment differ between NL, DE, and FR over 2021–2023? Which economic topic dominates each country's media narrative?

LEAD / LAG ANALYSIS
To what extent does media sentiment on economic topics lead or lag actual economic indicators (HICP, unemployment rate, house price index)? Does negative coverage predict deteriorating conditions, or does it merely reflect them?

DATA SOURCES

GDELT GKG
Global Knowledge Graph monitoring 185,000+ news sources in 100+ languages, updated every 15 minutes. Used to extract monthly article volume and sentiment scores per country and economic topic.

Eurostat HICP
Harmonised Index of Consumer Prices — monthly inflation rates per EU country broken down by category (food, energy, housing, services). Ground-truth inflation signal used in cross-correlation analysis.

Eurostat Labour & HPI
Country-level monthly unemployment rates and quarterly house price indices for DE, FR, and NL. Together with HICP, these cover all three research topics as verifiable ground-truth economic data.

METHODOLOGY

Data Integration & Analysis Pipeline

- R1 - COLLECTION**
Data Ingestion
GDELT GKG queried via BigQuery for articles matching country + topic keywords. Eurostat API pulled monthly HICP, unemployment, and house price data. Both stored in structured tabular format.
- R2 - INTEGRATION**
Knowledge Graph Construction
Entities (articles, months, countries, topics, indicators) modelled as nodes. Relations such as PUBLISHED_IN, COUNTRY, DISCUSSES, REPORTED_FOR linked via RDF-style triples in a Node.js graph store.
- R3 - CLEANING**
Preprocessing
Sentiment scores aggregated to monthly level. Outliers reviewed. Economic series aligned to identical monthly frequency for comparability.
- R4 - EXTRACTION**
Cypher Queries & CSV Export
Relevant knowledge graph queried using Cypher to retrieve aggregated monthly sentiment scores and economic indicator values per country/topic pair. Results exported as structured CSV files for downstream analysis.
- R5 - ANALYSIS**
Python Analysis & Visualisation
Expanded Data Loader in Python (pandas, matplotlib). Time series plots, stationarity checks, and Pearson cross-correlation analysis performed to answer RQ1 and RQ2. Results visualised as the charts presented in this poster.

KNOWLEDGE GRAPH

Graph Structure Overview

The knowledge graph integrates media articles, economic indicators, countries, months, and topics as interconnected nodes. Relations encode how articles discuss topics, in which country and month they were published, and how economic indicators are reported for each country-period pair.

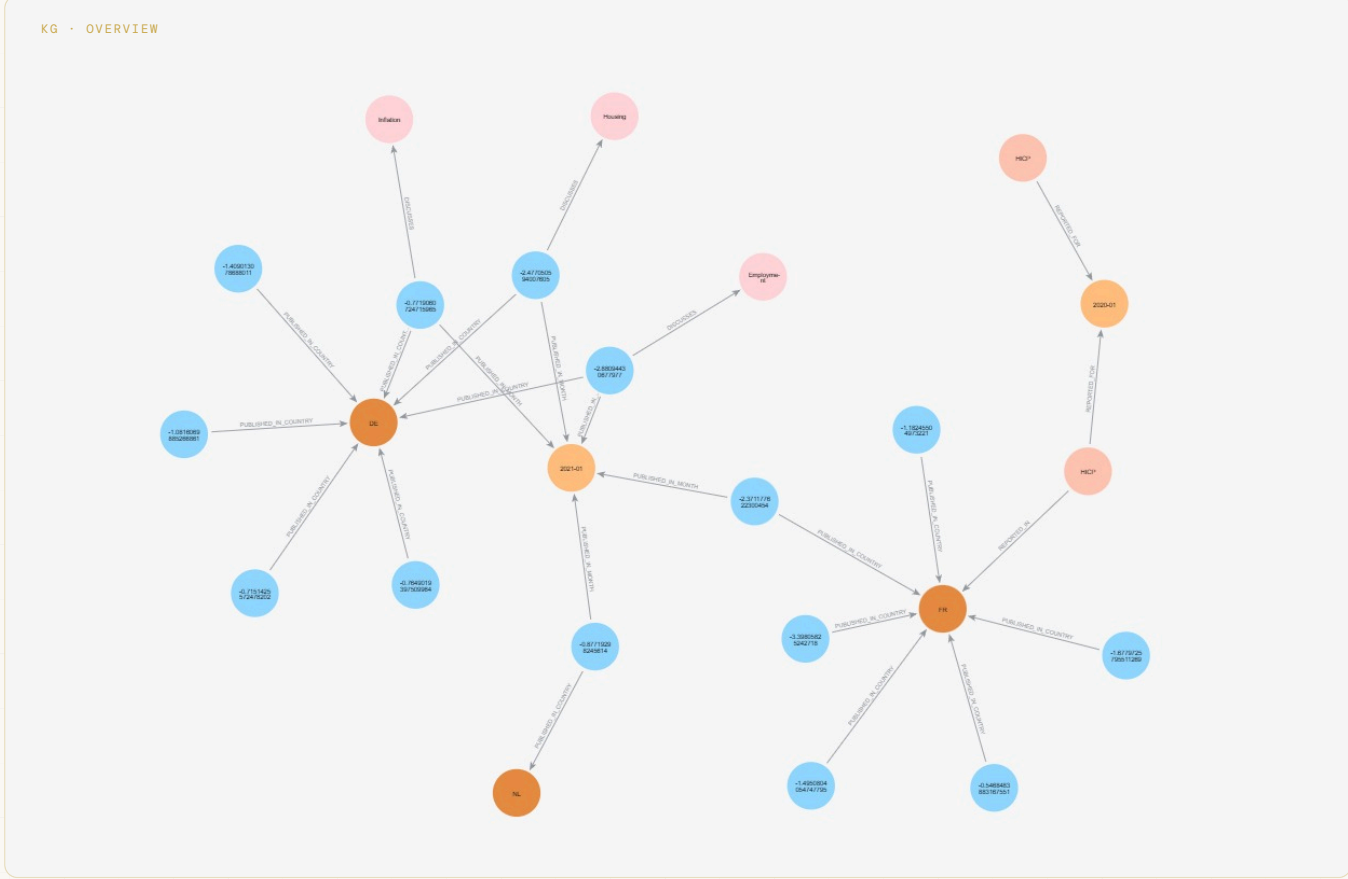


FIG KG-A - Full graph overview: all countries, topics, months, and selected article/indicator nodes

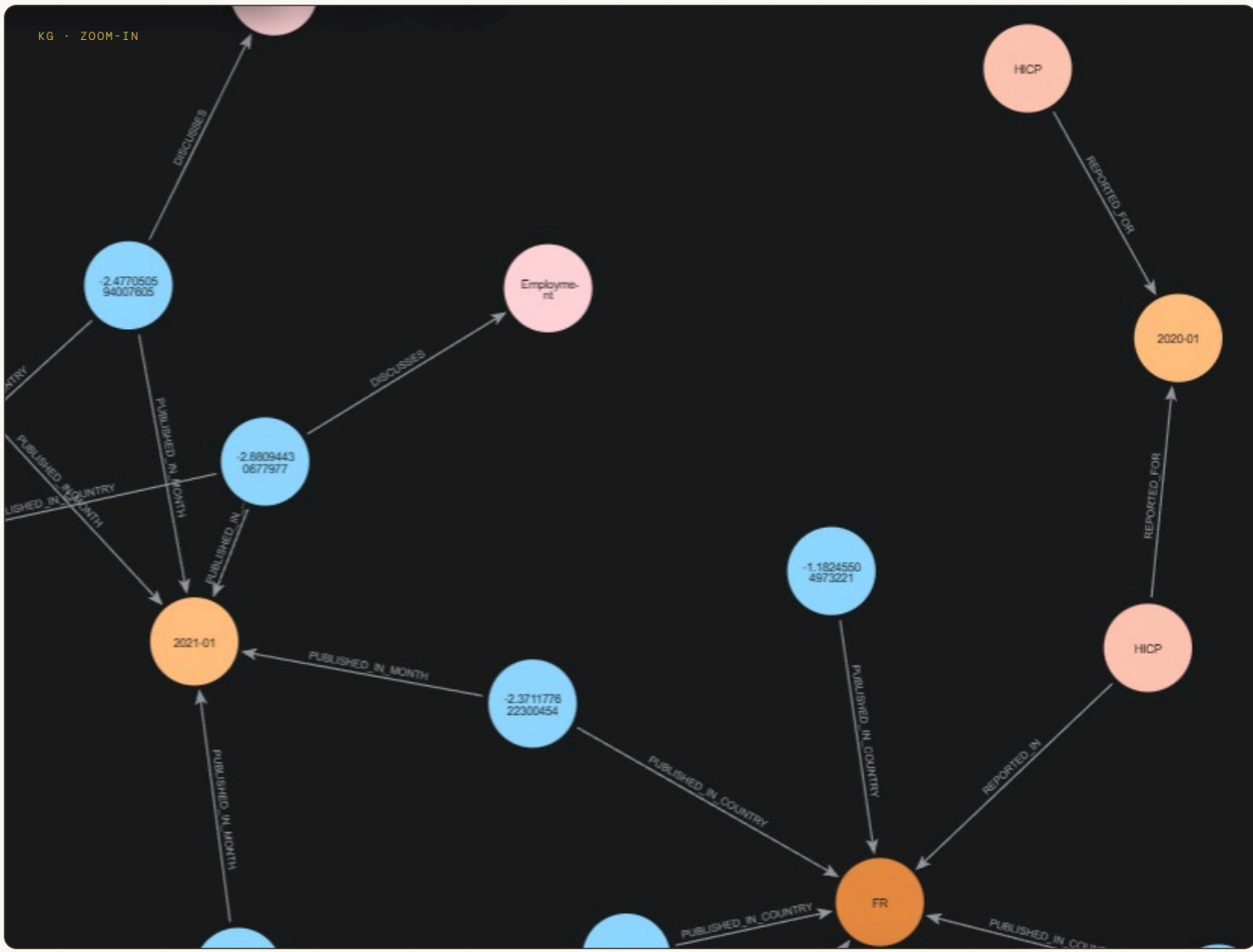


FIG KG-B - Detailed view: article nodes with sentiment scores, PUBLISHED_IN and DISCUSSES relations

RQ1 & RQ2 - SENTIMENT

Sentiment Trajectory Over Time

All three countries maintain predominantly negative sentiment on economic topics throughout 2021–2023. Employment sentiment is most volatile for NL (extreme dips in late 2021). Inflation sentiment in France exhibits sustained negativity across all lags, consistent with the cross-correlation findings.

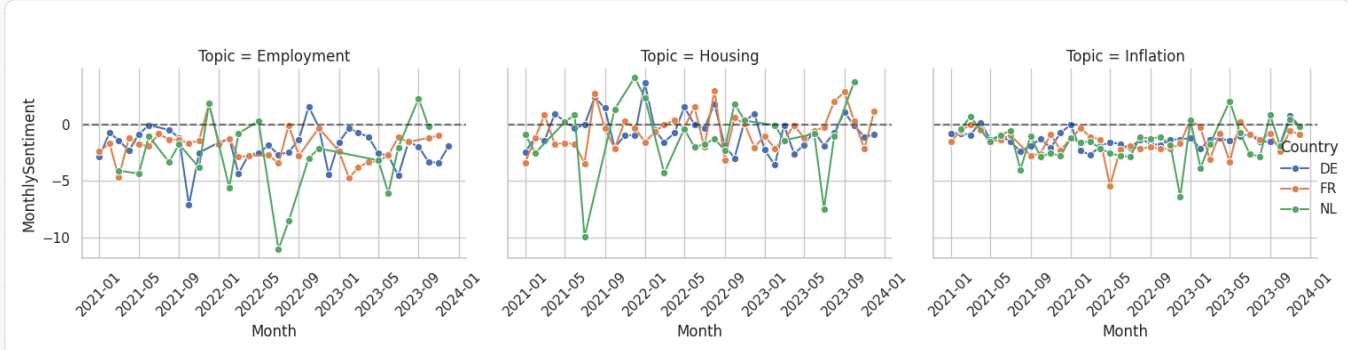


FIG 3 - Monthly aggregated sentiment score | Source: GDELT GKG

RQ1 - VOLUME

Total Media Volume by Topic & Country

Inflation dominates media coverage in all three markets, with Germany producing nearly 3x more inflation articles than the Netherlands. Employment receives the least coverage proportionally. This disparity reflects the 2022 energy-price shock which drove intense European media attention to consumer prices.

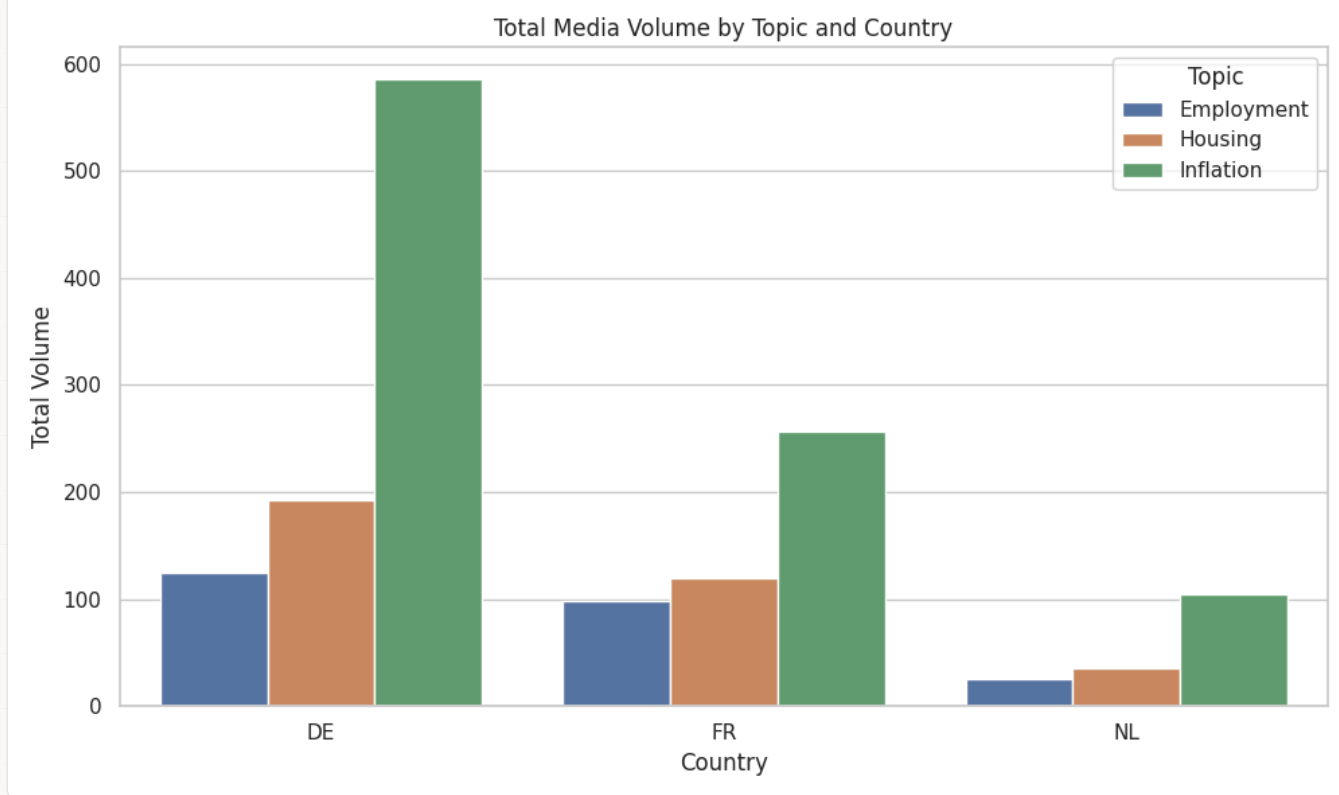


FIG 1 - Total article count 2021–2023 | Source: GDELT GKG

~590
DE INFLATION ARTICLES

~255
FR INFLATION ARTICLES

~105
NL INFLATION ARTICLES

RQ2 - TEMPORAL

Monthly Coverage Intensity

Media volume on inflation spikes sharply for Germany in mid-2022 (peaking at ~40 articles/month), coinciding with record eurozone CPI figures. Housing and employment coverage remain comparatively flat throughout the period, suggesting less editorial sensitivity to those indicators.

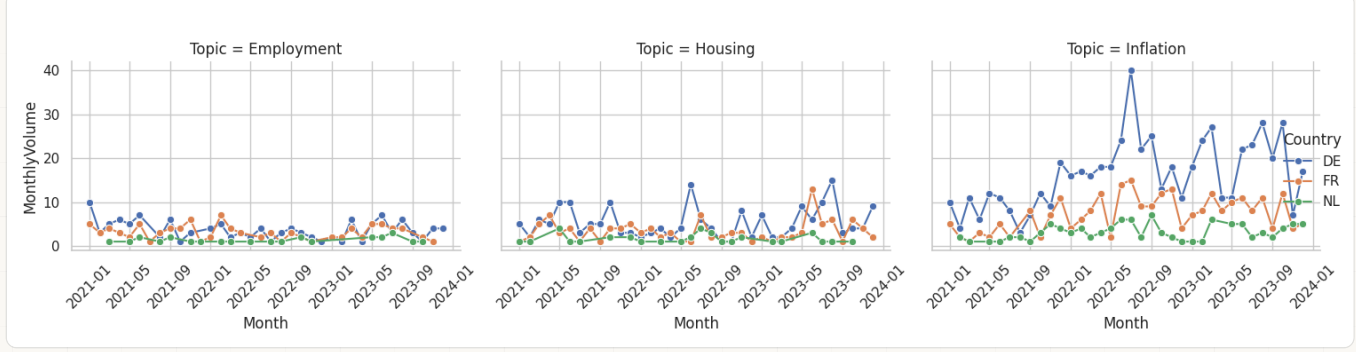


FIG 2 - Monthly article volume per topic/country | Source: GDELT GKG

METHODOLOGY - PREPROCESSING

Stationarity & Transformations

Media sentiment series (FIG 4) oscillate around a stable mean without persistent trends — confirming stationarity and validating direct use in cross-correlation. NL shows the highest variance, reflecting a smaller article pool.

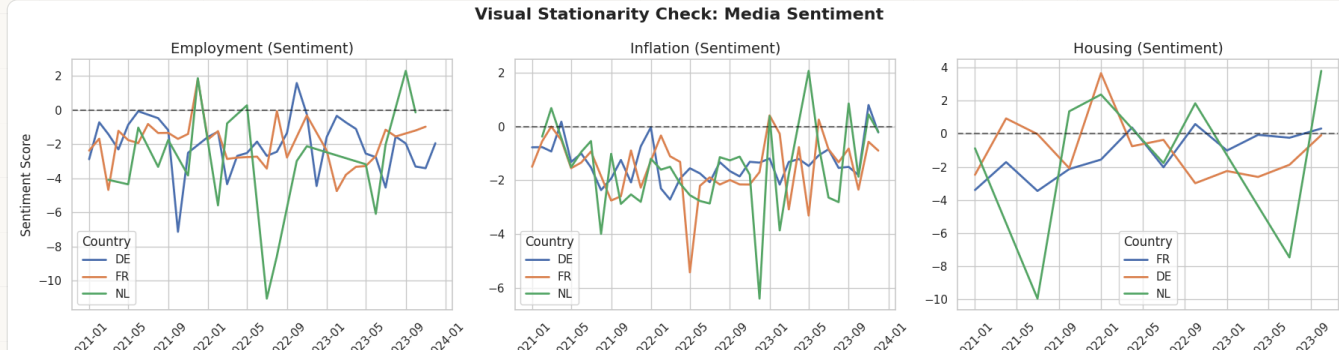


FIG 4 - Stationarity check: media sentiment scores 2021–2023

Economic indicator series (FIG 5), however, showed clear non-stationarity — most notably inflation, which exhibits a strong upward trend across all three countries. To satisfy the stationarity assumption required for cross-correlation analysis, two transformations were applied depending on the series:

DIFFERENCING First-order differencing ($\Delta y_t = y_t - y_{t-1}$) was applied to remove linear trends from unemployment and house price series.

LOG DIFFERENCING Log differencing ($\Delta \ln y_t$) was applied to inflation (HICP) to stabilise both trend and variance, yielding an approximate month-on-month growth rate.

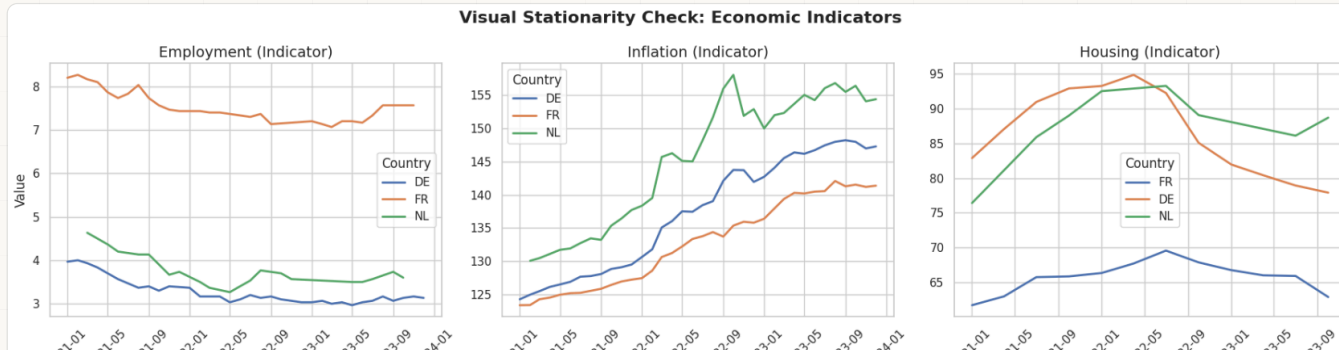


FIG 5 - Visual stationarity check: economic indicators 2021–2023 (pre-transformation)

RQ2 - LEAD/LAG ANALYSIS

Cross-Correlation: Sentiment vs. Economic Indicators

Pearson cross-correlations at ±3-month lags reveal a striking finding: **FR inflation sentiment shows strong negative correlations** ($r \approx -0.5$ to -0.8) at almost all lags, suggesting French media pessimism consistently co-moves with or leads rising inflation. DE and NL exhibit weaker, mostly sub-threshold correlations for employment and housing. NL housing sentiment shows significant negative correlations across all lags, indicating structural pessimism rather than a predictive signal.

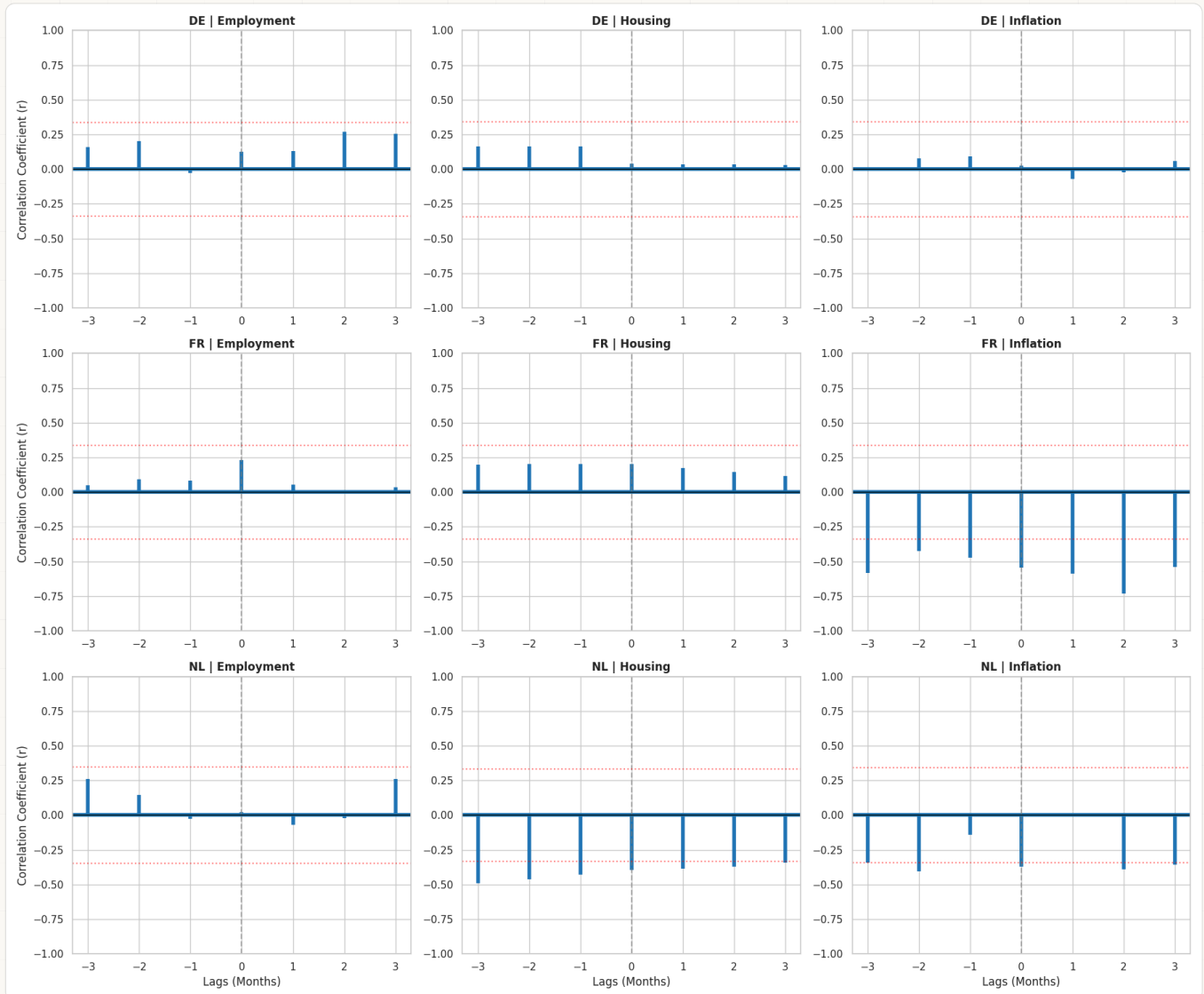


FIG 6 - Pearson r at lags ±3 months; red dashed lines = significance threshold ($r = \pm 0.3$)

● Correlation bar ● Significance threshold (± 0.3) — Lag = 0 (concurrent)

KEY FINDINGS

Answers to the Research Questions

Inflation Dominates Across All Markets

Inflation is by far the most covered economic topic in DE, FR, and NL media between 2021 and 2023. Germany produces the highest absolute volume, likely due to its larger media ecosystem and heightened public concern over energy price inflation. Employment receives least coverage proportionally, even though labour market dynamics directly affect insurance claim volumes.

DE & NL: Weak Leading Indicators

French inflation media sentiment correlates strongly with HICP movements. Integrate this signal into a monthly risk dashboard. When the rolling 3-month average sentiment score drops below a defined threshold, trigger a review of income protection and business interruption pricing in the French market before official statistics confirm deterioration.

FR Inflation Sentiment Strongly Predictive

France is the standout case: media sentiment on inflation is significantly and negatively correlated with the HICP indicator across all measured lags (±3 months). This implies that French media pessimism on inflation either co-moves with or marginally precedes deteriorating price conditions — a potentially actionable early warning signal for the insurer.

NL: High Variance, Small Sample Effects

The Netherlands exhibits the most volatile sentiment series, with extreme monthly scores driven by a comparatively small article pool. This increases statistical noise and limits the reliability of lag analysis for NL. Expanding the GDELT query scope or supplementing with Dutch-language sources would improve robustness for RQ2.

STRATEGIC RECOMMENDATIONS FOR THE CLIENT

From Insight to Insurance Action

Deploy FR Sentiment as an Early-Warning Signal

French inflation media sentiment correlates strongly with HICP movements. Integrate this signal into a monthly risk dashboard. When the rolling 3-month average sentiment score drops below a defined threshold, trigger a review of income protection and business interruption pricing in the French market before official statistics confirm deterioration.

Monitor NL Housing Narrative for Claim Surges

NL housing sentiment is persistently and significantly negative across all lags. While not a leading indicator per se, this structural pessimism correlates with elevated claim risk in property lines. Consider adjusting reserve buffers for Dutch property insurance proactively, and track whether sentiment volatility increases ahead of quarterly HPI releases.

Strengthen NL Data Pipeline

The high variance in Dutch sentiment reflects a thin article sample. Supplement GDELT with dedicated Dutch-language source scraping (e.g. NRC Handelsblad, De Telegraaf APIs) to improve monthly estimate stability. A larger NL corpus would likely surface statistically significant correlations for employment and inflation topics that are currently obscured by noise.

Automate Monthly KG Refresh

GDELT updates every 15 minutes. Build an automated pipeline that ingests new GKG records monthly, updates sentiment aggregations, appends new nodes and relationships to the knowledge graph, and re-runs cross-correlation analysis. This transforms the current retrospective analysis into a live operational intelligence feed aligned with the insurer's pricing review cycle.

Integrate into Pricing Model

Combine the sentiment index with the HICP, unemployment, and HPI economic indicators as composite features in a predictive pricing model. Media sentiment adds a "soft" dimension — consumer perception and fear — that official statistics alone cannot capture. Annuity incorporating both dimensions is better positioned to anticipate claim volume surges and adjust premiums or risk appetite proactively.

"Media narrative is not merely a reflection of economic reality — in some markets and for certain topics, it shapes the expectations that drive insurance claims. This analysis demonstrates that a knowledge graph approach, integrating open media intelligence with official economic indicators, can provide the insurer with a structured, scalable, and timely early-warning system."

RESEARCH CONCLUSION